



WYDZIAŁ ELEKTRONIKI
I TECHNIK INFORMACYJNYCH



Hierarchical Specialised Ensembles for Zebrafish Phenotype Classification

Piotr S. Maciąg Monika Maciąg Magdalena Majdan

Warsaw University of Technology Nencki Institute Medical Universities of Lublin and Warsaw

KES 2026

Outline

- 1 Introduction
- 2 Hierarchical ensemble designs
- 3 Experimental results
- 4 Conclusions

Introduction

Problem statement

- Zebrafish embryos are widely used to study toxicological and pharmacological effects.
- Manual assessment of microscopy images is slow, subjective and sensitive to observer differences.
- Several defects can appear in the same embryo, so a single-label classifier is not enough.

Question

Whether creating specialised ensembles of classifiers can improve current results in zebrafish embryo recognition and classification?

The task is both hierarchical and multi-label. The model must first identify a broad state, then describe co-occurring defects.

The dataset mixes exclusive and overlapping states

Dataset context

Experiments use the public zebrafish embryo dataset introduced by Jeanray et al. (2015), containing microscopy images annotated with 11 phenotype labels.

Stage 1: one state per image

Chorion Dead Normal or *Other*

742 / 186 / 512

training / validation / test images

Stage 2: defects may co-occur

- **Lesions:** Edema, Hemostasis, Necrosed Yolk Sac
- **Tail and body morphology:** Down Curved Tail, Short Tail, Up Curved Fish, Up Curved Tail, Up Curved TailFish

Representative zebrafish phenotypes

Representative zebrafish phenotypes — Training Set



Chorion



Dead



Normal



Down Curved Tail • Edema • Hemostasis • Necrosed Yolk Sac • Short Tail

Edema • Necrosed Yolk Sac • Up Curved Tail • Up Curved TailFish



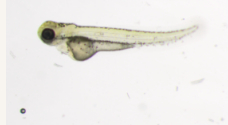
Hemostasis • Edema • Necrosed Yolk Sac • Up Curved Fish • Up Curved TailFish



Necrosed Yolk Sac • Up Curved Fish • Up Curved TailFish



Short Tail • Edema



Up Curved Fish • Up Curved TailFish



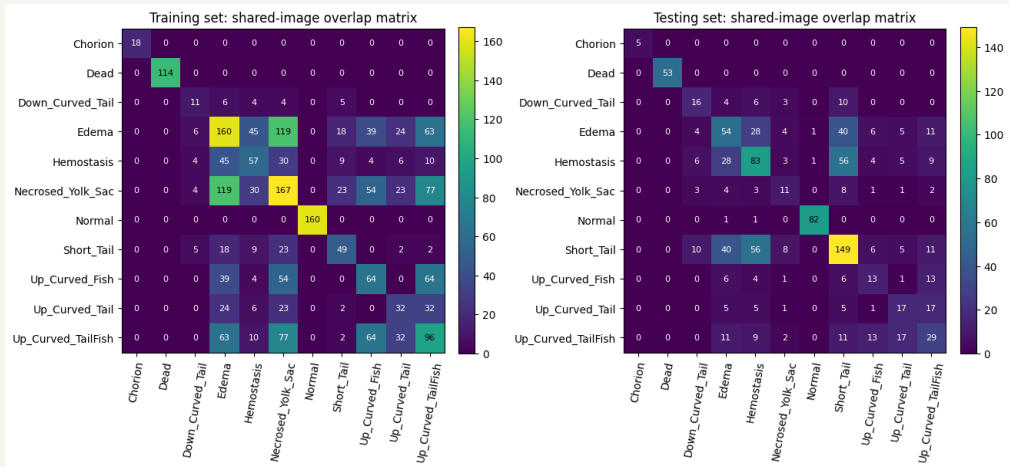
Up Curved Tail • Edema • Hemostasis • Up Curved TailFish



Up Curved TailFish • Edema • Necrosed Yolk Sac • Up Curved Fish



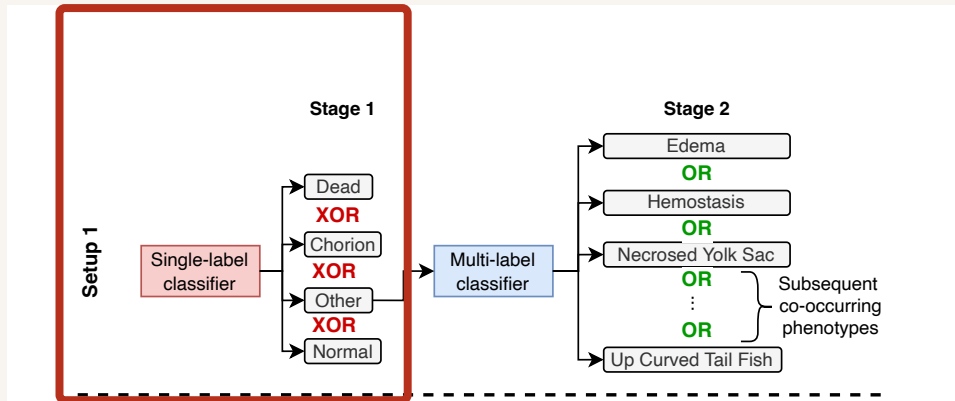
Co-occurrence is not a corner case



The off-diagonal counts confirm substantial label overlap.

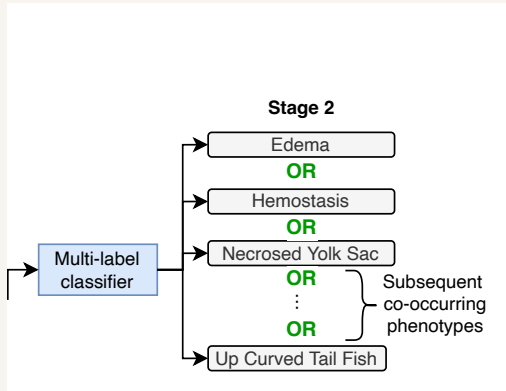
Hierarchical ensemble designs

Every setup starts with the same four-way gate



- Stage 1 selects exactly one state: Dead, Chorion, Normal or Other.
- Only images assigned to Other continue to multi-label analysis.

Setup 1: one model for all co-occurring defects



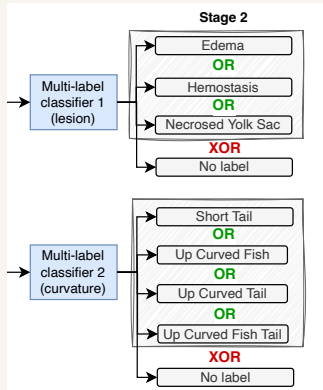
One shared model

One multi-label classifier predicts all eight stage-two defects.

The model learns a common representation for lesion and shape-related labels.

The model shares evidence across defects, but must learn several visually different phenotypes.

Setup 2: two specialists for co-occurring defects



Lesion expert

Edema, Hemostasis and Necrosed Yolk Sac.

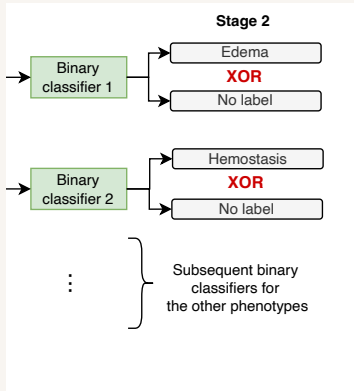
Shape expert

Short Tail and the upward-curvature defects.

Each expert may return no positive label.

Related defects share a model, while each specialist handles a narrower phenotype family.

Setup 3: one classifier per co-occurring defect



Eight binary decisions

Each classifier predicts whether one assigned defect is present or absent.

The eight outputs form the final multi-label prediction.

Each decision is highly specialised, but the classifiers cannot share representations across defects.

The comparison holds the training recipe constant

Backbones

- ResNet18
- ViT tiny, patch size 16
- ConvNeXt tiny

All models use pretrained weights from `timm`; the same backbone is used in both stages of a run.

Training and decisions

- 224×224 inputs; batch size 16
- 10 epochs per stage; learning rate 10^{-4}
- weight decay 10^{-4}
- per-label thresholds selected on validation data from 0.05 to 0.95 to maximise F1

Architecture and ensemble design are compared under the same optimisation settings and data splits.

Experimental results

ConvNeXt leads; setup 2 gives the best F1 balance

Setup 1

	Prec.	Rec.	F1	Acc.
R18	.32	.85	.38	.85
ViT	.33	.46	.32	.53
CNeXt	.40	.53	.42	.55

Setup 2

	Prec.	Rec.	F1	Acc.
R18	.35	.61	.40	.61
ViT	.38	.68	.45	.74
CNeXt	.40	.73	.47	.76

Setup 3

	Prec.	Rec.	F1	Acc.
R18	.33	.69	.41	.72
ViT	.34	.71	.41	.79
CNeXt	.36	.87	.44	.84

0.47

best mean F1: setup 2
+ ConvNeXt

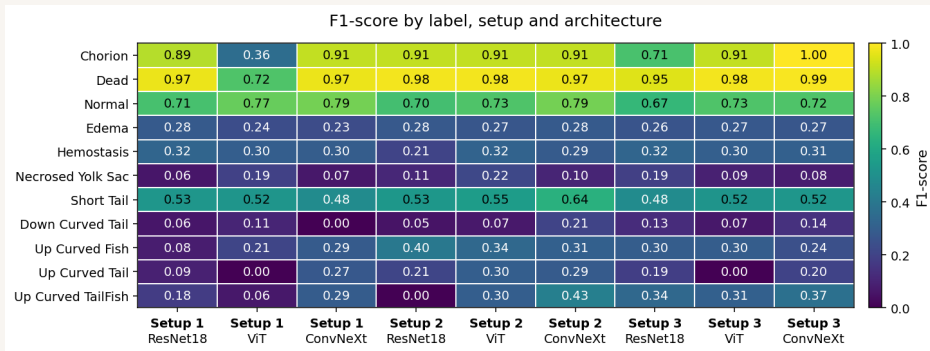
0.87

best mean recall: setup
3 + ConvNeXt

0.85

best reported accuracy:
setup 1 + ResNet18

Performance separates clean states from subtle defects



Dead is stable; Necrosed Yolk Sac and curved-tail labels remain hard.
Setup 2 with ConvNeXt is strongest across many labels.

Higher recall does not always mean a better balance

Setup 2 + ConvNeXt

Precision .40 Recall .73 F1 .47

The specialised pair of multi-label models gives the best mean F1-score.

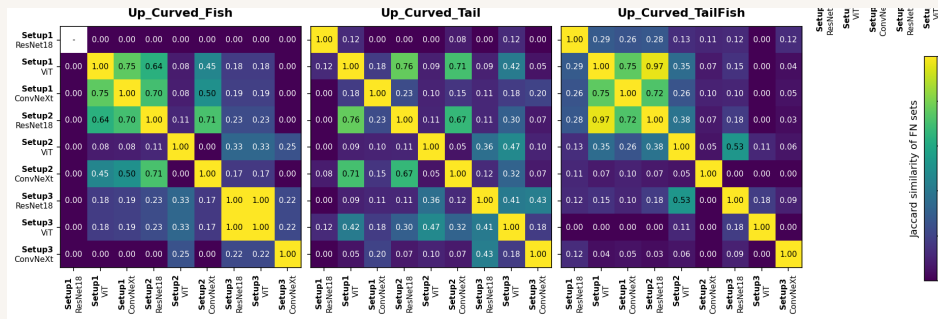
Setup 3 + ConvNeXt

Precision .36 Recall .87 F1 .44

Independent binary models find more positives, but also produce more false positives.

The binary ensemble is attractive when recall dominates. For a balanced classification objective, setup 2 performs better.

Different models do not always miss the same embryos



False-negative overlap varies by phenotype and model. Higher similarities occur more often in setup 3, suggesting that its binary classifiers repeatedly miss some of the same difficult samples.

Conclusions

1. **ConvNeXt is the strongest backbone overall**, including on several subtle morphology labels.
2. **Setup 2 has the best mean F1-score**: two phenotype-family models retain useful shared structure.
3. **Setup 3 maximises recall**, but its lower precision reduces the overall balance.

A hierarchy helps, but the useful level of specialisation sits between one broad multi-label model and a separate classifier for every defect.

The next gains may come from explicit morphology

Quantify curvature

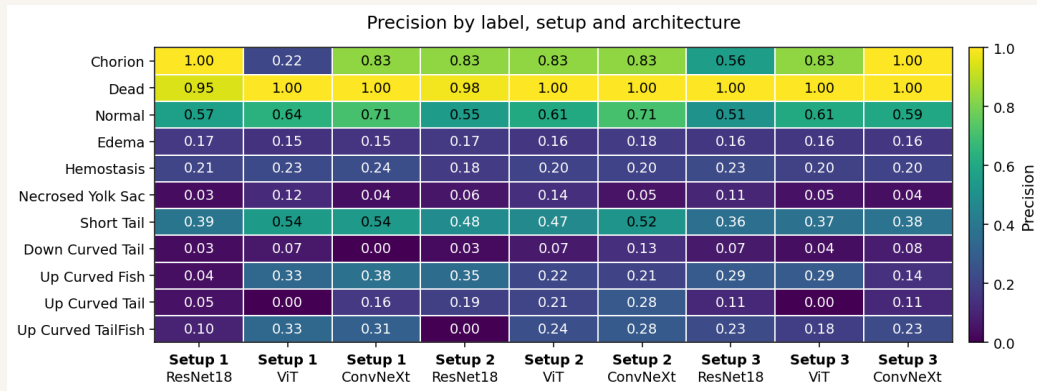
Tail length and curvature measures could help distinguish subtle upward- and downward-curved phenotypes.

Describe more anatomy

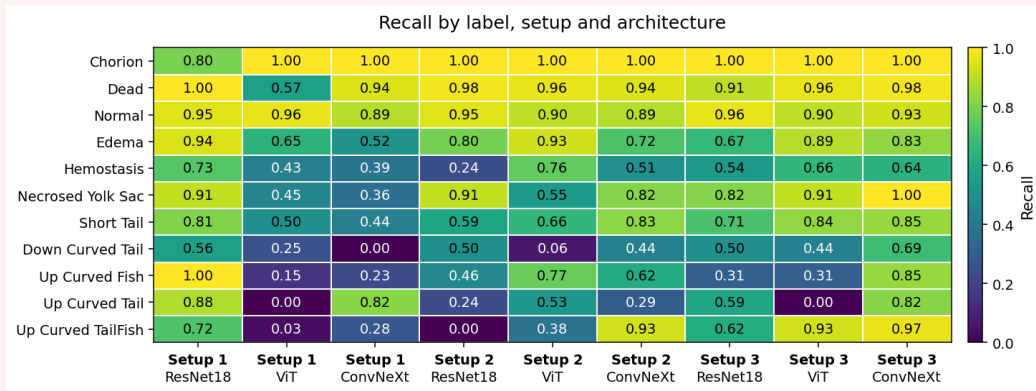
Eye size, yolk-sac edema and internal-organ malformations could add evidence beyond the image-level label.

The hardest labels are morphological. Better descriptors may complement stronger backbones and ensemble design.

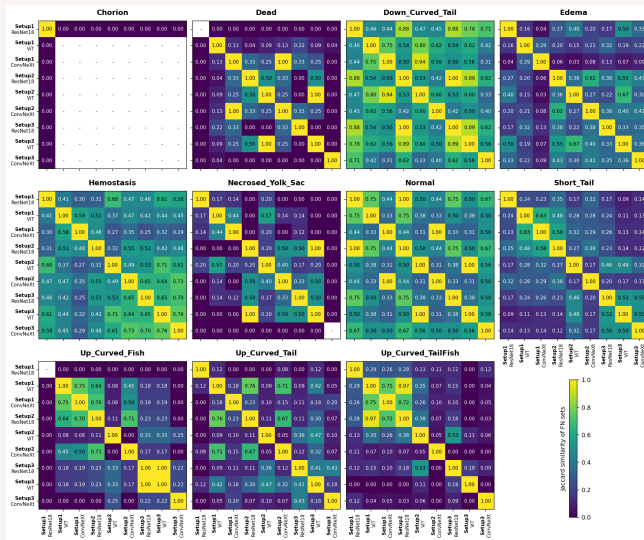
Backup: precision by phenotype



Backup: recall by phenotype



Backup: complete false-negative overlap analysis



Backup: evaluation measures

$$\text{precision}(c) = \frac{TP(c)}{TP(c) + FP(c)},$$

$$\text{recall}(c) = \frac{TP(c)}{TP(c) + FN(c)},$$

$$F1(c) = \frac{2 \text{ precision}(c) \text{ recall}(c)}{\text{precision}(c) + \text{recall}(c)},$$

$$J(A, B) = \frac{|A \cap B|}{|A \cup B|}.$$

Metrics are averaged across phenotype classes for the aggregate comparison. Jaccard similarity is computed between false-negative sets.

Selected references

- Jeanray, N. et al. (2015). Phenotype classification of zebrafish embryos by supervised learning. *PLOS ONE*, 10(1), e0116989.
- Liu, Z. et al. (2022). A ConvNet for the 2020s. *CVPR*, 11976–11986.
- Puchalla, J., Serianni, A., and Deng, B. (2025). Zebrafish identification with deep CNN and ViT architectures using a rolling training window. *Scientific Reports*, 15, 8580.
- Tandon, A. et al. (2025). AI-driven morphological assessment of zebrafish larvae for developmental toxicity chemical screening. *Aquatic Toxicology*, 285, 107415.

All figures and experimental values in this deck are taken from the supplied article source and its accompanying files.